appear distinct only at low energies and temperatures. For a more detailed discussion see, for example, *The Elegant Universe,* Chapter 7.

2. Once we know that a field, like any of the known force fields, is an ingredient in the makeup of the cosmos, then we know that it exists everywhere—it is stitched into the fabric of the cosmos. It is impossible to excise the field, much as it is impossible to excise space itself. The nearest we can come to eliminating a field's presence, therefore, is to have it take on a value that minimizes its energy. For force fields, like the electromagnetic force, that value is zero, as discussed in the text. For fields like the inflaton or the standard-model Higgs field (which, for simplicity, we do not consider here), that value can be some nonzero number that depends on the field's precise potential energy shape, as we discussed in Chapters 9 and 10. As mentioned in the text, to keep the discussion streamlined we are only explicitly discussing quantum fluctuations of fields whose lowest energy state is achieved when their value is zero, although fluctuations associated with Higgs or inflaton fields require no modification of our conclusions.

3. Actually, the mathematically inclined reader should note that the uncertainty principle dictates that energy fluctuations are inversely proportional to the time resolution of our measurements, so the finer the time resolution with which we examine a field's energy, the more wildly the field will undulate.

4. In this experiment, Lamoreaux verified the Casimir force in a modified setup involving the attraction between a spherical lens and a quartz plate. More recently, Gianni Carugno, Roberto Onofrio, and their collaborators at the University of Padova have undertaken the more difficult experiment involving the original Casimir framework of two parallel plates. (Keeping the plates perfectly parallel is quite an experimental challenge.) So far, they have confirmed Casimir's predictions to a level of 15 percent.

5. In retrospect, these insights also show that if Einstein had not introduced the cosmological constant in 1917, quantum physicists would have introduced their own version a few decades later. As you will recall, the cosmological constant was an energy Einstein envisioned suffusing all of space, but whose origin he—and modern-day proponents of a